Leveraging CSI of WiFi for Breath Estimation: An RNN-Based Rate Computation Using ESP32 Data

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***Abstract*—Wireless vital sensing faces challenges due to a lack of non-intrusive, cost-effective solutions for long-term health monitoring. Existing research on Channel State Information (CSI) for vital sign monitoring reveals gaps, particularly in standardized datasets and the use of widely available hardware. This paper addresses these gaps by proposing a novel method that utilizes CSI data for breathing rate estimation with the cost-effective ESP32 WiFi device. A comprehensive CSI dataset was collected for breathing rates from 6 to 30 Breaths Per Minute (BPM), and a specialized Recurrent Neural Network (RNN) model was developed to estimate these rates accurately. The model demonstrated high accuracy, with a Mean Absolute Error (MAE) of 0.04 and an Absolute Percentage Error (APE) of 0.027. Comparative analysis with existing datasets confirms the robustness of this approach. This study highlights the potential for a practical, cost-effective, and non-intrusive solution for long-term health monitoring using existing WiFi infrastructure.**

***Index Terms*—WiFi sensing, Channel State Information, Feature Extraction, Breathing Rate Estimation, RNN**

I. Introduction

The increasing prevalence of chronic health conditions such as lung disorders and heart disease has heightened the need for cost-effective, non-intrusive health monitoring solutions suitable for long-term indoor use. Traditional methods, such as capnometers and pulse oximeters, are often cumbersome and impractical for continuous monitoring. Wireless technology has opened new avenues for unobtrusive health monitoring, with non-contact methods such as cameras, radar, and acoustic signals showing promise [1]. However, these approaches face challenges such as lighting sensitivity, high hardware costs, and complexity of implementation. For example, camera-based methods require adequate lighting and may not be suitable for environments such as sleep monitoring, while acoustic systems are limited by range and user capacity.

Recent advancements in Radio Frequency (RF) systems, particularly RADAR-based technologies [2], have shown exceptional accuracy in detecting vital signs. However, these systems often involve high costs and require specialized hardware. WiFi-based systems [1] leveraging CSI, have emerged as a cost-effective alternative due to their widespread availability. CSI offers detailed insights into wireless channel interactions, providing a more accurate and stable basis for monitoring physiological signals than Received Signal Strength (RSS). Furthermore, various machine learning techniques have been applied, including Convolutional Neural Networks (CNN), Artificial Neural Networks (ANN), and ensemble methods [3]–[8] to monitor vital signs using CSI data. These approaches have effectively analyzed CSI data and accurately estimated breathing rates. CNNs detect spatial features, while ANNs handle complex data relationships, improving precision in identifying breathing rates. Ensemble methods further enhance reliability by combining model outputs, reducing errors.

Despite these advancements, challenges remain, including the reliance on customized hardware, the lack of standardized datasets, and the under-explored potential of neural networks, particularly RNNs, in combination with widely available WiFi hardware like the ESP32. Addressing these gaps is essential for developing practical and widely applicable vital sign monitoring solutions.

This paper proposes an approach leveraging CSI data for breathing rate estimation using commodity WiFi devices. The main contributions are:

- **WiFi CSI Data Collection**: Captured a dataset of WiFi CSI data using ESP32 micro-controller covering breathing rates from 6 to 30 BPM and ground-truth for varying breathing rates using a self-designed breath controlling software. The "ground truth for varying breathing rates" provides actual, verified breathing data as a benchmark, allowing precise comparison with the CSI data.
- **Development of Neural Network**: An RNN model is designed specifically to estimate breathing rates from CSI data. The model incorporates advanced preprocessing techniques and hyper-parameter tuning to enhance its accuracy and robustness.
- **Evaluation**: The developed model has been evaluated with both our dataset and publicly available datasets.

II. Background Work

Integrating WiFi technology into vital sign monitoring has introduced a non-invasive method for remotely tracking physiological parameters. Utilizing CSI, which captures changes in WiFi signal propagation, researchers have successfully monitored diaphragmatic breathing patterns. During inhalation, the

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Paper/work	Device Used	Technology Used	Methodology	Preprocessing	Approach	Regression
[2]	TI 77 GH FMCW mmWave	RADAR	Machine Learning	Dopler FFT	LSTM	✓
Ubibreathe, [9]	Mobile and any AP	RSS	Peak Detection	Bandpass Filter	FFT	✓
TensorBeat, [10]	5 GHz Commditity WiFi	CSI	Peak Detection	Decomposition	Autocorrelation	✓
PhaseBeat, [11]	Intel 5300 NIC	CSI	Peak Detection	Subcarrier Selection	DWT	✓
[12]	Intel 5300 NIC	CSI	Peak Detection	Savitzky–Golay filter	PCA+VBA	✓
MoBreath, [13]	Smartphone (Nexus 5)	CSI	Peak Detection	Segmentation+DWT	FFT	✓
[14]	Intel AX200 WiFi Card	CSI	Peak Detection	Hampel Filter	FFT	✓
[15]	Intel 5300 NIC	CSI	Machine Learning	PSD peak	K-means	✓
[3]	TL-WDR4300 router	CSI	Deep Learning	Hampel Filter	CNN	✓
[4]	ZTE ZXHN H298A router	CSI	Deep Learning	Domain Shifting	ANN	✓
[16]	ESP32	CSI	Peak Detection	Filtering & PCA	SPSD Peak	✓
[5]	ESP32	CSI	Machine Learning	Decomposition	ResDT	×
Proposed work	**ESP32**	**CSI**	**Deep Learning**	**Scaling**	**Simple RNN**	✓

TABLE I: Comparison of Existing Solutions

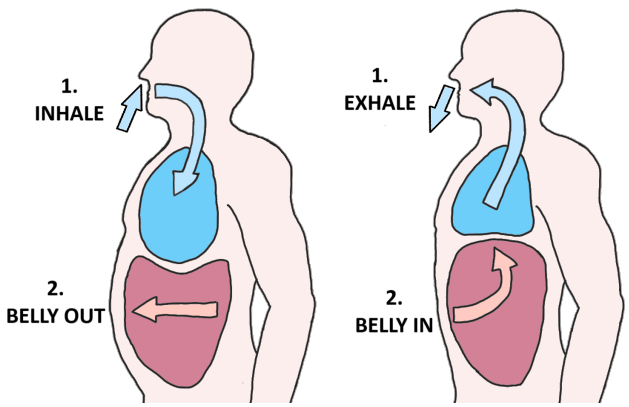

Fig. 1: Diaphragmatic Breathing

diaphragm's downward movement causes abdominal expansion, while exhalation results in contraction, as illustrated in Fig. 1. These movements create subtle variations in CSI, enabling the detection of breathing phases. By analyzing the interactions between WiFi signals and the human body, including reflections and diffractions, CSI-based methods provide a promising tool for vital sign monitoring with significant potential for healthcare applications.

When a WiFi device, such as a router or access point, transmits a signal, it sends an RF wave that interacts with objects and people in its path. These interactions cause reflections, refractions, and diffractions, altering the signal. A receiver, such as a WiFi adapter or smartphone, detects this modified signal and measures changes in amplitude and phase, represented as complex values in the CSI matrix H. The received signal can be expressed as:

$$Y = Hx + \epsilon$$

where Y is the received signal, x is the transmitted signal, H is the CSI matrix, and ϵ represents noise. This equation encapsulates the essence of CSI-based monitoring.

The CSI matrix H is multidimensional, describing channel characteristics between each pair of transmit and receive antennas over multiple subcarriers [1], [10], [11]. Its dimensions are defined by the number of transmit antennas (N_t), receive antennas (N_r), and subcarriers (N_s) giving H, the dimensions $N_r \times N_t \times N_s$. To illustrate, consider different subcarrier counts. In the case of 52 subcarriers, as specified in the IEEE 802.11a/g/n standards, the WiFi signal is divided into 52 subcarriers. The CSI matrix H captures the channel characteristics for each subcarrier. For instance, with 2 transmit antennas and 3 receive antennas, H will have dimensions $3 \times 2 \times 52$, meaning for each of the 52 subcarriers, there is a 3×2 matrix detailing the channel information. For advanced standards like IEEE 802.11ac, which use up to 234 subcarriers, the CSI matrix H provides even finer frequency resolution. With the same number of transmitting and receiving antennas, H would have dimensions $3 \times 2 \times 234$. This higher subcarrier count enhances environmental monitoring accuracy by offering detailed channel characteristics. The number of subcarriers directly impacts the resolution of the CSI data, with a higher number providing finer granularity for monitoring. As movements affect the WiFi signal's propagation, they cause subtle CSI changes. By analyzing these variations, researchers can extract vital sign information, such as breathing and heart rates, allowing non-invasive healthcare applications.

III. STATE-OF-THE-ART

Numerous studies explore CSI for vital sign monitoring. For instance, TensorBeat [10] uses tensor decomposition to estimate breathing rates of up to five individuals with over 75% accuracy for errors under 2 bpm, addressing the limitations of previous methods relying on RSS. By analyzing CSI phase difference data, TensorBeat demonstrates the potential of RF sensing for health monitoring. Another system, PhaseBeat [11], monitors respiration and breathing rates using CSI phase differences from commodity WiFi. Employing techniques like Orthogonal Frequency Division Multiplexing (OFDM) and Discrete Wavelet Transform (DWT), PhaseBeat makes significant contributions despite some limitations like high median error for breathing rate estimation and the need for detailed signal processing knowledge. In [14], Intel AX200 with 2 Tx and 2 Rx antennas is used to capture one minute of CSI data across 245 subcarriers at 5.57 GHz with 20 MHz

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bandwidth. The best Rx-Tx pair is selected, and the CSI signal is processed using filters like Hampel and Butterworth for a specific subcarrier. Fast Fourier Transform (FFT) is applied to the processed CSI amplitude to estimate breathing rates via peak detection.

Building on this foundation, studies have also explored deep learning techniques for human biometric signals and contactless respiration monitoring. [2] used long short-term memory (LSTM) to exploit a correlation between the movement intensity and vital signs to estimate vital signs. In [3], researchers used CNNs to detect breathing activity from CSI data, achieving accurate breathing rate estimation due to CNNs' hierarchical feature extraction capabilities. However, the high computational complexity of CNNs poses challenges for real-time processing on resource-limited devices. Also, in [4], ANNs were employed to model complex relationships within CSI data for breathing rate estimation. Although ANNs provide flexibility in handling complex datasets, they may face challenges in capturing the subtle temporal and spatial relationships within CSI data, resulting in accuracy limitations that hinder their effectiveness.

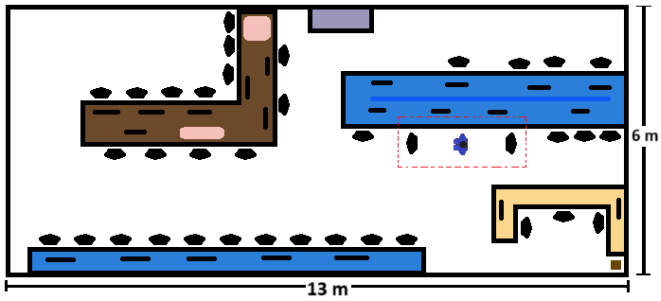
Synthesis: While many studies rely on specific hardware, [5] tackled the cost-effectiveness issue by proposing a digital twin model for respiration monitoring using the affordable ESP32. The model uses WiFi CSI, signal processing, and machine learning to classify respiratory status (e.g., normal, tachypneic, bradypneic). Though promising, it treats breathing rates as categorical variables, potentially missing subtle pattern changes. This highlights the need for regression-based methods for more precise respiratory monitoring, which this study aims to address. Table I shows comparisons with the different state-of-the-art approaches.

IV. Proposed Methodology

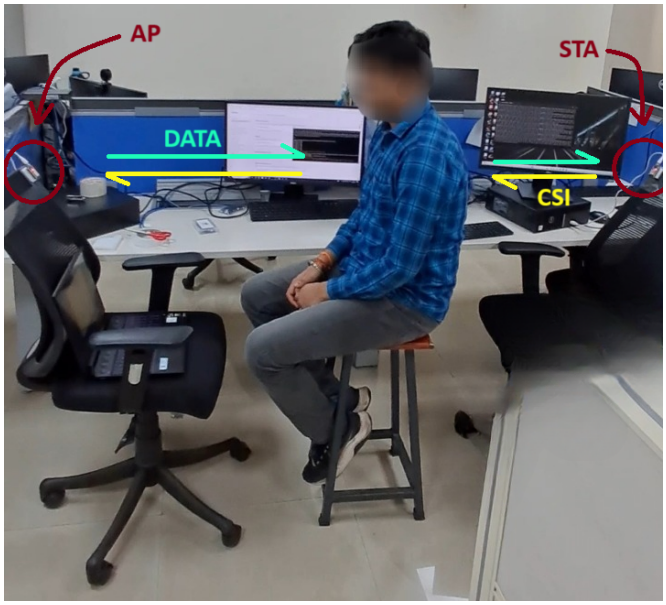
The proposed solution is modular in design, comprising several integrated modules that work together to achieve accurate breathing rate estimation. The proposed solution is represented in Fig. 5. The following subsections provide a detailed overview of each module:

A. Experimental setup

For the experiment, two WROOM ESP-32U Development Boards were utilized, leveraging their built-in RX and TX functionalities. These modules support the IEEE 802.11n standard. They operate within a frequency range of 5,180 MHz to 5,825 MHz and offer channel bandwidths of 20 MHz, 40 MHz, and 80 MHz. The data rate can reach 867 Mbps with OFDM modulation, supporting up to 228 subcarriers in an 80 MHz channel. Each module features an onboard PCB antenna with a transmit power of 27 ± 3 dBm and a receiver sensitivity of -82 dBm. Fig. 2a illustrates the laboratory setup. Data collection occurred in a 6×13 meter lab, intentionally preserving the typical environment to simulate real-world conditions. Essential appliances like an air conditioner, router, computer, and workstation were kept operational to introduce potential interference. This approach assessed environmental impacts on breathing data accuracy, replicating practical application scenarios more closely. Two ESP32 modules were chosen to establish a controlled transmitter/receiver setup as proofs-of-concept, allowing precise monitoring of CSI data between them. This setup is ideal for our experiment as it enables clear signal transmission and reception within a defined channel, providing reliable data on environmental changes. Using two ESP32s simplifies the initial analysis, and this configuration can easily be expanded with additional modules for more complex testing scenarios.



(a) Experiment Lab



(b) Closeup of Experimental setup

Fig. 2: Data Collection Environment

B. Data Collection

Fig. 2b provides a close-up of the experimental setup, where two ESP32 modules are placed about 2 meters apart. Using the ESP32-CSI tool [17], CSI information is captured continuously during data transmission, with the baud rate set at 1.5×10^6, yielding a sample rate of 78 packets per second. To simulate breathing rates, a custom Graphic User Interface (GUI) prompts the user to inhale and exhale at specified BPM rates while also logging timestamps for respective BPM for later analysis. As shown in Fig. 3, an orange ball guides

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breathing: it prompts inhalation when entering a blue zone and exhalation in a white zone, marking one breath cycle.

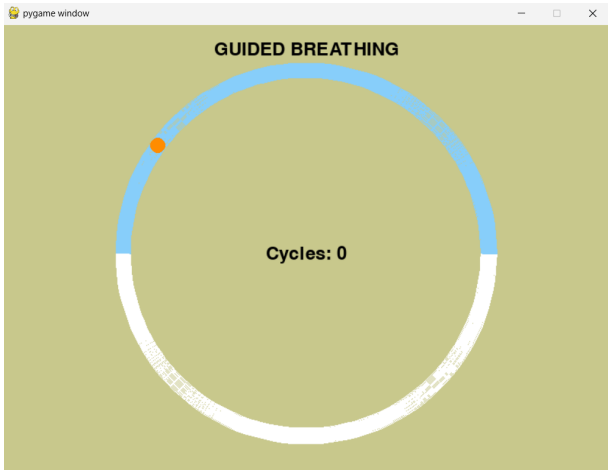
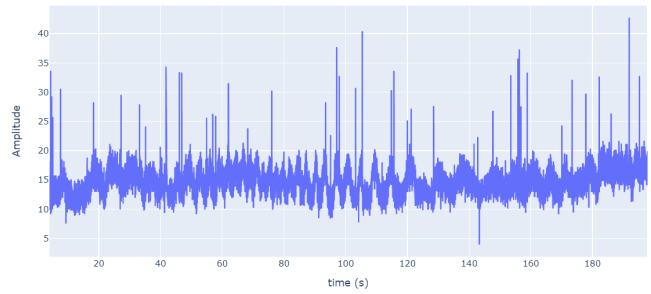


Fig. 3: Breath Controlling Aid

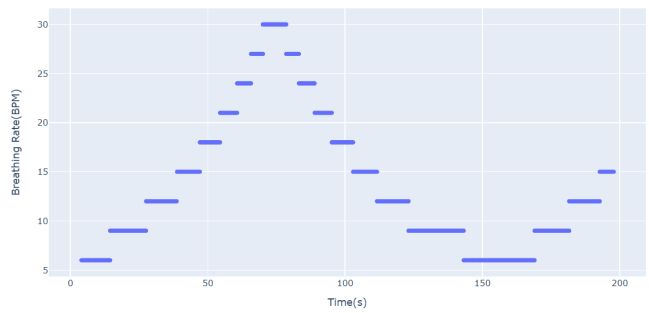
A 21-year-old male subject (180 cm, 74 kg) sat between the sensors, as shown in Fig. 2b. One ESP32 served as the transmitter, the other as the receiver. To capture the breathing data, the subject focused on the GUI to precisely regulate his breathing rate over a duration of 1.5 minutes, utilizing 192 subcarriers to extract detailed features from the WiFi CSI data. This was repeated at increasing BPM rates from 6 to 30 in 3 BPM intervals. Additional data for variable rates (6 to 30 BPM and back down) was collected over 3 minutes, alongside ground truth data. Data was stored in a `.csv` file with 26 columns, many holding consistent values. For instance, `type` always contained `"CSI_DATA"`, and `role` indicated `"AP"`. Fixed entries included `rate` at 11 Mbps, `sig_mode`, `bandwidth`, `smoothing`, `not_sounding`, and `aggregation` at 1 (showing OFDM, wide bandwidth, signal smoothing, active transmission, and packet aggregation), while `stbc`, `fec_coding`, and `sgi` were 0. `Noise_floor` was -96 dBm, `channel` was set to 6, and `secondary_channel` to 1. `Ant` and `rx_state` were preset to 0, and `real_time_set` to 0, meaning the time reference started from zero. Variability appeared in columns such as `local_timestamp` (local time), `real_timestamp` (elapsed time), `rssi` (-77 to -64 dB), and `mcs` (0 to 6). `Ampdu_cnt` ranged from 0 to 6, reflecting packet aggregation count, and `sig_len` provided CSI symbol length, with `len` used to calculate subcarrier numbers. Finally, the `CSI_DATA` column stored CSI as a list, where odd indices represented the real parts, and even indices represented the imaginary parts. The CSI of the ith subcarrier was calculated as $2n+j(2n+1)$.

C. Preprocessing

After data collection, significant impurities were found, requiring thorough cleaning. Hence, the process first entailed filtering out incomplete rows and consistently structuring the data to include 26 features with CSI data of length 384. A third filter ensured MAC address, signal length, and CSI length



(a) CSI amplitude of single subcarrier



(b) Distribution of Ground-Truth

Fig. 4: Visualization of Collected Data

consistency. Data captured within the first 5 seconds were discarded to remove initial noise and unstable values caused by sensor calibration and setup settling time. This period often includes environmental adjustments and transient signals that could skew the CSI readings, ensuring only stable, reliable data is used for accurate breath estimation, and columns with identical values across all rows were removed, resulting in 8 distinct features: `rssi`, `mcs`, `aggregation`, `ampdu_cnt`, `local_timestamp`, `sig_len`, `real_timestamp`, and `CSI_DATA`. These filtered features extracted complex-valued CSI data and separated into 384 columns for all subcarriers' real and imaginary parts (1 to 192). The pilot symbol locations were identified in the subcarrier numbers 1-6, 33, 60-66, 124-134, and 192. Subsequently, 166 columns containing the amplitude values of CSI data for each subcarrier were created. Finally, the original 384 columns, along with `mcs`, `aggregation`, `sig_len`, and `local_timestamp`, were dropped, resulting in a dataset with 169 features. The extracted amplitude is shown in Fig. 4a, and the ground truth distribution is illustrated in Fig. 4b.

D. Model creation

Given the temporal and spatial nature of the data, RNN was chosen over CNNs(which are more suited for only spatial features) for training due to its capability to handle sequential data and capture temporal dependencies through recurrent connections. Data were scaled using a MinMax Scaler before training. This scaling improves model convergence during training, leading to faster learning and potentially better accuracy, especially for models sensitive to input magnitudes like neural

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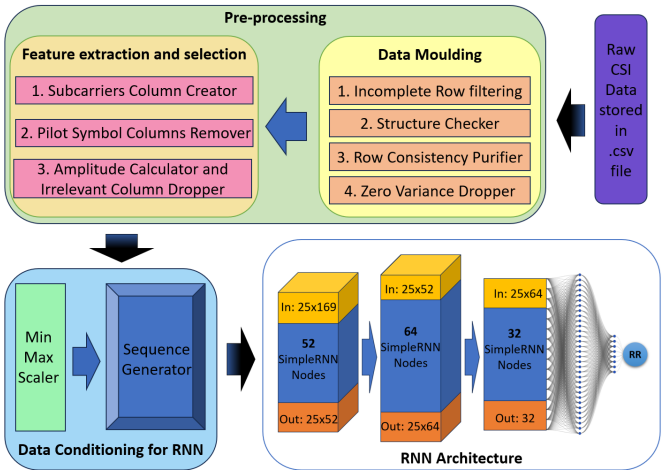

Fig. 5: Proposed Architecture

networks. The 2-dimensional DataFrame was transformed into a 3D temporal sequence of shape $((T \times s) - t) \times t \times f$, where T is the time elapsed in seconds since data capturing, s is the sampling rate, t is the timestep size, and f is the number of features. In this case, the sequence shape was $14534 \times 25 \times 169$. The target vector was similarly transformed to shape $(T \times s) - t$.

Components	Details
Input	14534 × 25 × 169 (inputs × batch_size × features)
RNN Blocks	3 Simple RNN layers with units 52, 64, and 32
FC Layers	3 layers with neurons: 64, 16, and 1
Output	1 (respiration_rate)
Trainable Params	25,305 (size: 98.85 KB)

TABLE II: RNN Architecture Specifications

The model specifications are provided in Table II, and hyperparameter tuning was applied to optimize performance. For all RNN layers, `go_backwards` was set to `True` to capture long-term dependencies, and `return_sequences` to `True` for the first two layers. Used the `relu` activation function throughout, as the target was numerical and trained for 350 epochs with a batch size of 25, using the `adam` optimizer and `mse` as the loss metric.

V. Result and Discussion

Multiple RNN models were trained using CSI data, with some using raw data and others with preprocessed data involving outlier removal and frequency domain conversion. As in [4], a model trained on amplitudes scaled using the RSSI values achieved an APE of 4.42%. The vanilla RNN effectively captured temporal features without domain conversion. To simplify and reduce processing time, necessary features were scaled and fed directly into the model. The timestep was set to 25, given the sampling rate of approximately 75 Hz, meaning each timestep corresponds to about 1/3 of a second. This approach effectively captured temporal relationships within each second of data. Cross-validation was used for hyperparameter tuning and to prevent overfitting, with some data reserved for testing. This approach enabled effective analysis of CSI data for breathing rate detection using RNNs.

For model evaluation, its performance is tested on data collected at fixed breathing rates, following initial training on a 3-minute sample of variable breathing rates. The same preprocessing as [4] technique is applied to these datasets, and small, randomly selected portions of the processed data are clipped. When fed into the model, the predicted breathing rates closely match the labels provided for the constant breathing rate data, demonstrating the model's accuracy. Finally, the model's overall performance is compared with other related works, as shown in Table III. Few studies specifically address the regression task of breathing estimation, as most reshape the problem into a classification task. This work adheres to the regression approach and compares its regression metrics with available studies.

Paper/work	MSE	MAE	APE	Method
Proposed work	0.051944	0.043278	0.002794	Scaling + RNN
[4]	0.025	NaN	2.7	Proper Preprocessing + ANN
[16]	1.081	0.434	NaN	DWT + PCA + PSD peak
[18]	0.014	NaN	NaN	Peak detection + spectrum filtration

TABLE III: Comparison of different methods

The only similar publicly accessible dataset is described in [14]. This dataset, collected with an Intel AX200 WiFi card, contains CSI data from two transmitting and two receiving antennas across eight experiments. Two of these experiments involve two individuals, and six involve a single person. Among the single-person experiments, three were conducted at constant breathing rates (12 BPM, 15 BPM, and 20 BPM), while the remaining three involved variable breathing rates. Ground-truth data were obtained using a respiration belt, but fluctuations led to the selection of samples with stable ground-truth values. The Intel AX200 captures CSI data for 245 subcarriers, but only 169 were used for model processing to avoid emphasizing periodic signal patterns, considering the subtle influence of breathing in the training data.

The model is validated by comparing its results with the [14] dataset. The evaluation mirrors the original study's methodology, extracting subsets with nearly constant breathing rates from the variable breathing rate data. Mean Squared Error (MSE) and MAE are calculated for these samples, and the Maximum Repeated Predicted Value (Mode) is also considered to assess performance. As shown in Fig. 6, the errors of the proposed model are either close to or lower than those obtained using the peak detection method from the reference dataset.

The average metrics for different breathing rates across

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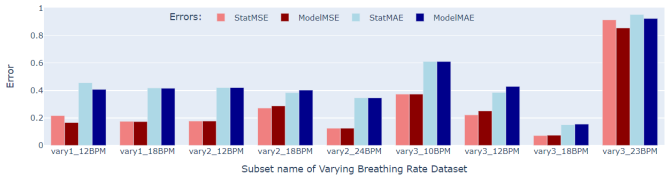

Fig. 6: Error Comparison with varying Datasets

the three variable-rate datasets were calculated for a comprehensive comparison. For the constant-rate datasets, random-duration CSI data samples from the entire dataset were input into the network, and regression metrics were computed. As shown in Fig. 7, the model's estimations closely align with the ground truth, as the error margins indicate.

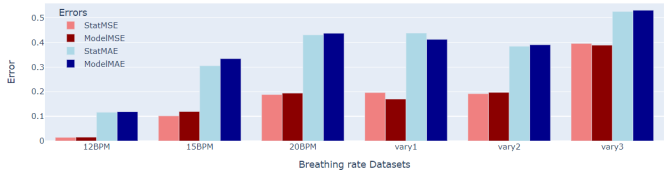

Fig. 7: Error Comparison with Available Dataset

However, the model's errors were slightly higher in some instances than peak estimations. This discrepancy may be due to the quality of the dataset used for training. The dataset was collected with low-cost hardware in a practical environment with numerous active devices, whereas the comparison dataset was collected using a WiFi card under near-ideal conditions with multiple antennas, resulting in slightly better performance.

VI. Conclusion and Future Scope

This paper showcases the potential of using CSI data from commodity WiFi devices to accurately estimate breathing rates. The study achieved high accuracy in breathing rate estimation by developing a specialized RNN model and compiling a comprehensive dataset. Notably, a simple RNN model proved to be an effective alternative to traditional frequency domain conversion and peak detection methods, outperforming existing approaches. Future research directions include exploring ensemble learning for model development, implementing real-time processing, conducting clinical validation, enhancing system robustness against environmental and hardware variations, developing algorithms for multi-person breathing rate estimation, and exploring the use of CSI data for heart rate estimation. These efforts will expand the capabilities of CSI-based vital sign monitoring and contribute to creating comprehensive, contactless healthcare solutions.

VII. Acknowledgment

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